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Storytelling Dashboard – Part 2

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Declaration of Originality

I am the originator of this work and I have appropriately acknowledged all other original sources used as my references for this work.

I understand that Plagiarism is the act of taking and using the whole or any part of another person's work, including work generated by AI, and presenting it as my own.

I understand that Plagiarism is an academic offence and if I am found to have committed or abetted the offence of plagiarism in relation to this submitted work, disciplinary action will be enforced.

Task 1 : Visual Cues

Prompt: My dataset contains values like channel (phone and branch), date, time, resolution status, waiting time, feedback score, latitude longitude and name of road. Recommend me the best chart type for specific variables.

Gen AI response: It recommended to use box plot for channel vs waiting time, scatter plot for feedback score vs waiting time, line chart or heatmap for time-of-day 1vs feedback, point density map for location vs feedback, line chart for date vs number of resolved cases. For individual variables, bar chart for channel, feedback score, road name and resolution status, line chart for date and time, histogram for waiting time, filled map or point map for latitude/longitude.

Here you go — clear, specific visualisation recommendations for each variable in your dataset. This is perfect for your *Visual Cues* section (Part 2) and fits well into a 1–2-page explanation.

✓ Best Chart Types for Each Variable in Your Dataset

Below are the recommended visualisation techniques based on whether the variable is categorical, time-based, numerical, or geographical.

1. Channel (Phone / Branch)

Type: Categorical (2 categories)

✓ Best Visualisations:

- Bar Chart (Vertical or Horizontal)
 - Best for comparing counts of "Phone" vs "Branch".
- Pie Chart / Donut Chart (only because there are 2 categories)
 - Useful if proportions are important (e.g., 70% phone, 30% branch).
- Stacked Bar Chart (if combined with another variable such as resolution or satisfaction)

2. Date

Type: Time-series (daily/monthly)

✓ Best Visualisations:

- Line Chart
 - Best for trends over time (e.g., number of customers per day).
- Area Chart
 - If you want to show cumulative or volume effect.
- Heatmap (Calendar Heatmap)
 - Shows which days of the week/month have higher activity.

3. Time (Time of Day)

Type: Continuous time (HH:MM)

✓ Best Visualisations:

- Line Chart (frequency by hour)
 - Shows peak periods (e.g., busiest hours).
- Histogram / Density Plot
 - Shows distribution of service times.
- Circular/Radial Clock Plot (advanced)
 - Best if your data has strong patterns across the day (24-hour cycle).



★ Bonus: Best Multi-Variable Visual Insights

If you want to impress your lecturer with deeper insights:

1. Channel vs Waiting Time

- → Use Box Plot or Violin Plot

2. Feedback Score vs Waiting Time

- → Use Scatter Plot with trendline

3. Time of Day vs Feedback

- → Use Line Chart or Heatmap

4. Location (Lat/Long) vs Feedback/Issues

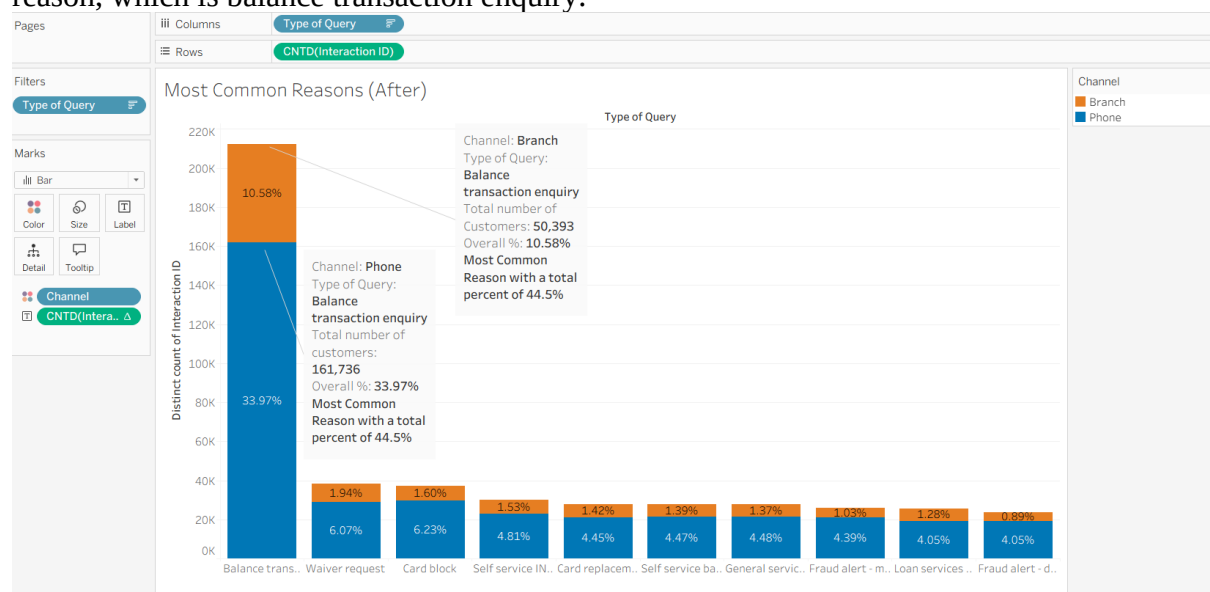
- → Use Point Density Map

5. Date vs Number of Resolved Cases

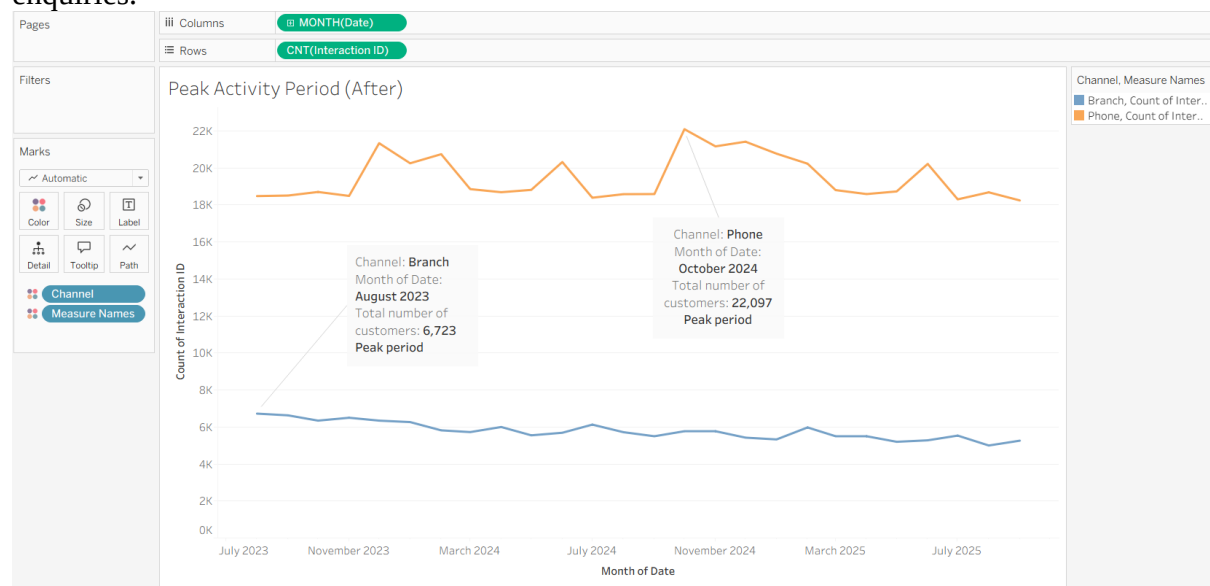
- → Use Line Chart for trend analysis

Implementation: I chose 5 graphs best relating to the problem statement and did not want to overwhelm by choosing too many visualizations. Bar chart was the most appropriate for channel and enquiry comparison. Latitude and longitude was also used to show a feedback score visual in map for all branches.

Enhancement 1: I tried to declutter the graph by filtering to only the Top 10 reasons as the previous graph was very messy with a bunch of columns with bars that are very low and may not have meaningful impact. The graph was also filtered to descending order for better visuality and readability. I also added the percent of total for each reason to better view and compare the various reasons. Annotations were also done to tell us about the most common reason, which is balance transaction enquiry.



Enhancement 2: I used annotations to notify the peak period for Phone and Branch to better visualise exactly which month and the number of customers coming in or calling for enquiries.



Task 2: Dashboard

Prompt: When creating a dashboard for 4 visuals, recommend me some dashboard design elements like layout, interactivity etc

AI Response:

✓ Visual Zoom/Pan (for Maps)

If using latitude/longitude, allow:

- Pinch-to-zoom
- Hover information
- Highlighting hotspots

3. Visual Consistency (Colours, Fonts, and Style)

✓ Colour Palette

Use a simple palette (2–4 colours maximum):

- One primary colour (e.g., blue)
- One accent for highlights (e.g., orange or red)
- Neutral colours (grey tones) for background elements

✓ Consistent legends and labels

- Put legends either at the right or bottom (same place for all)
- Avoid overly large labels
- Use the same date formatting everywhere (e.g., DD/MM/YYYY)

✓ Font Style

Use ONE font family throughout:

- Title (18–22 px, bold)
- Axis labels (12–14 px)
- Data labels (10–12 px)

Clean = professional.

4. Use of Space

Your dashboard should feel balanced.

✓ Leave enough white space

Avoid clutter. Add margins between visuals so the viewer can breathe.

✓ Avoid overlapping roles

Each visual should answer a different question:

- Visual 1: Trend over time
- Visual 2: Channel comparison
- Visual 3: Waiting time distribution
- Visual 4: Location performance or feedback

This prevents redundancy.

✓ Recommended Dashboard Design Elements (For 4 Visuals)

1. Layout & Structure

A clean, consistent layout helps users understand your dashboard at a glance.

Best 4-visual layout patterns:

Option A: 2 × 2 Grid (Most Common & Balanced)

- Top-left: Key Trend (e.g., Date trend)
- Top-right: Category comparison (e.g., Channel)
- Bottom-left: Numeric distribution (e.g., Waiting time)
- Bottom-right: Map or Feedback score breakdown

Option B: Hierarchical (Top summary, bottom details)

- Top row (wider): Time-series/overall KPI
- Bottom row (two visuals side-by-side): Comparisons and distributions
- Bottom panel: Geographic/map or detailed breakdown

Option C: Focal Visual Layout

- Left column: 1 tall main visual (e.g., map or trend)
- Right column: 3 smaller visuals stacked

Choose based on your MOST IMPORTANT visual → place it top-left or large and on the left side (users read left → right).

2. Interactivity Features (Highly Recommended for Marks)

✓ Filters (Slicers)

Add 2–3 slicers placed at the top or left side:

- Channel (Phone / Branch)
- Date range
- Resolution status
- Feedback score

Keep slicers consistent (aligned vertically or horizontally).

✓ Cross-filtering Between Charts

Enable click interactions:

- Clicking on "Phone" filters all visuals
- Selecting a feedback score updates waiting time distributions
- Clicking on a road highlights its map position

This makes your dashboard feel dynamic and modern.

✓ Tooltips

Add tooltips so hovering shows extra info:

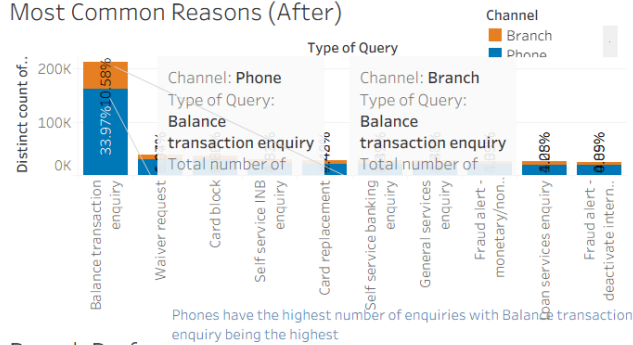
- Hover on a map point → road name, waiting time, feedback
- Hover on feedback bar → cases count, average waiting time

Tooltips increase insight without cluttering the dashboard.

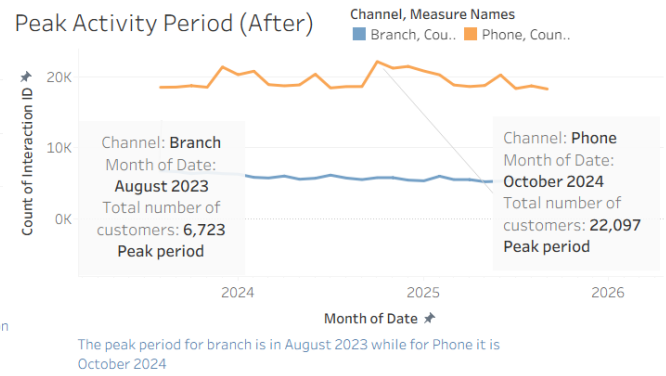
Implementation: I used the layout suggested by AI to maintain a clean and clear structure and easy to read left to right format. Having enough white space for my dashboard was crucial as I needed to write down my sticky notes for each visual, and to ensure that all my graphs are not cluttered together and have enough space for visuality. Using consistent colour for my branch and phone column was also useful to ensure any colour-blind concerns.

Dashboard

Most Common Reasons (After)

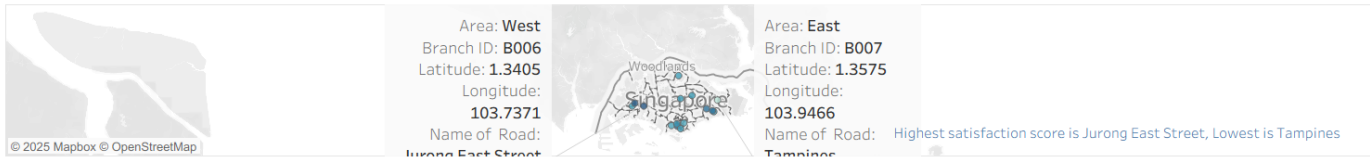


Peak Activity Period (After)

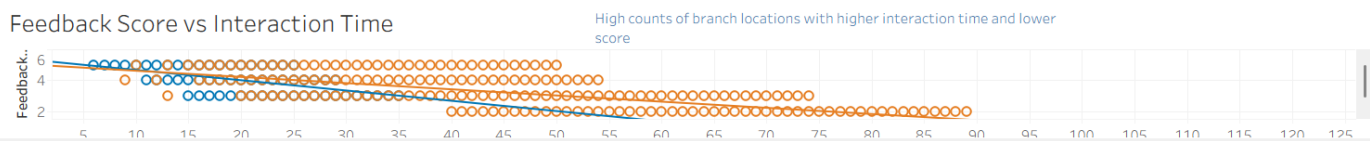


A lot of incoming calls for balance transaction enquiries with peak periods being October 2024. Branch in Jurong East Street has the highest s...

Branch Performance



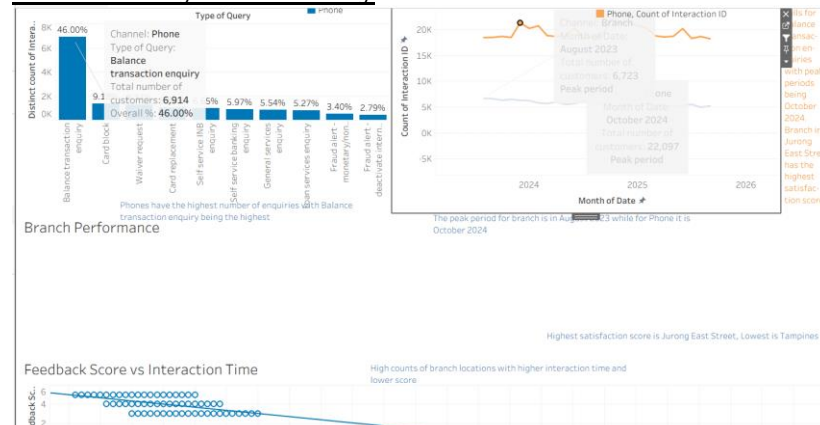
Feedback Score vs Interaction Time



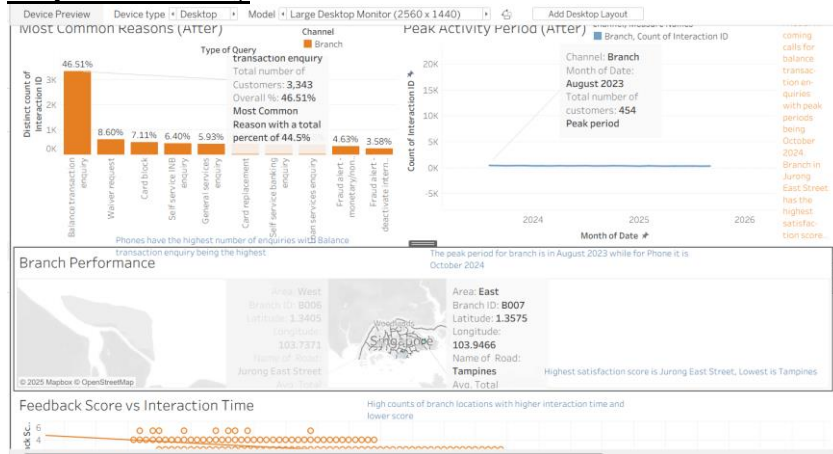
The top row follows 50-50 splits between most common reasons and peak activity period. A left to right reading will allow the reader to look at the most immediate root cause and move on to timely trends. The middle row shows branch performance followed by feedback score vs interaction time scatter plot. This provides a well-balanced layout and easy to read. Visual cues include consistent colours, orange and blue, for branch and phone with high contrast keeping in mind of colour-blind. Annotations highlighting the main point of each graph tells the reader more on what to focus and they do not have to stare at it and find it themselves. For hierarchy, white space is around visuals to avoid any cluttering. The dashboard provides a clear and effective communication. Titles are descriptive, visual complement text, overall, it avoids any confusion.

For interactivity, one of them is being able to use the peak activity period and find out the percentage of each reason at that point of time or month for both the branch and phone columns. The map also acts as a filter. For example, if we click on a branch dot in Tampines, it will update the bar, line and scatter to show only the associated queries and feedback score as well as the trend line for peak activity. This will allow the user to get more specific insights and can identify trends even better. When clicking on the phone for peak activity period, the map is expected to be blank as it is only supposed to show the branch locations, not phone. The examples are shown below:

Peak Activity Interactivity



Map Interactivity



Task 4 : Big Idea, Linear Storyboard, Narrative Framework

Prompt: Now, recommend me storyboarding to communicate data and insights, provide me with a brief introduction, highlighting key findings and clear recommendations to address issues such as heavy walk-in enquiries etc for OCBC bank

AI Response: The Gen AI gave me a storyboard with recommendations such as virtual queue, appointment system etc. These can address heavy walk ins. I used this as inspiration and was able to help me plan my own tableau story clearly.

congestion.

🔥 Storyboard Flow (6-Step Visual Narrative)

🔥 Frame 1 — Customer Activity Overview

Purpose: Provide a high-level snapshot of service volume.

Suggested Visual: Line chart (number of enquiries over time) + KPI cards.

Key Finding (example):

- Walk-in enquiries increased significantly during peak periods (10 AM–1 PM).
- Phone enquiries remain stable throughout the day.

Message: Demand is uneven, with branches absorbing the majority of peak traffic.

🔥 Frame 2 — Channel Comparison (Phone vs Branch)

Purpose: Identify which channel contributes to operational strain.

Suggested Visual: Bar chart / 100% stacked bar chart.

Key Finding:

- Branch channel accounts for a higher share of enquiries compared to phone.
- Walk-in customers show higher waiting times and more pending cases.

Message: Heavy walk-ins create service load and delay resolution progress.

🔥 Frame 3 — Waiting Time Analysis

Purpose: Show where service bottlenecks occur.

Suggested Visual: Box plot or histogram of waiting time (phone vs branch).

Key Findings:

- Median waiting time for branch customers is significantly higher.
- Outliers suggest some customers wait excessively long (over 20–30 mins).
- Phone channel is more consistent with lower variability.

Message: Branch queues are long and unpredictable, reducing service satisfaction.

🔥 Frame 4 — Customer Satisfaction & Resolution Status

Purpose: Link operational issues to customer perception.

Suggested Visual: Bar chart (feedback score distribution), stacked bar (resolution by channel).

Key Findings:

- Lower feedback scores are associated with long  branch waiting times.
- Pending and escalated cases occur more often at branches.

Big Idea

What's the Big Idea - Template

From the book *Storytelling with Data* by Cole Nussbaumer Knaflic.

Who's the Audience?

List primary group or individuals
If narrow down to one, who?
What do they care about?

OCBC
Head of Call Center

What's at stake?

What are the benefits?
What are the risks?
What's the urgency?
Action needed from audience

Call abandoned rate will remain and increasing, waiting time will remain and increase, lesser satisfied customers

Improve customer experience and reduce the overall waiting time, happier
Prevent customers from leaving to competing banks
Approve and fund the seasonal staffing and retraining

What's the Big Idea

In one sentence

Transform OCBC overwhelmed services by proper staffing and retraining underperforming branches

The first two parts are essentially the "So-what" Test, and the last part is for developing the Big Idea.

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The audience can be the head of both call centre operations and head of customer experience. They care about improving customer experience and to reach the target of 4.2 CSAT. They are motivated to stay ahead of their competitors such as DBS and to hit the feedback score target. They need to approve the proposal for self-service, seasonal staffing and retraining branches.

Linear Storyboard

Storyboard

Create a storyboard using the sticky notes.

PROBLEM

Phones have the highest number of enquiries with Balance transaction enquiry being the highest

The peak period for branch is in August 2023 while for Phone it is October 2024

High counts of branch locations with higher interaction time and lower score

ACTION/ ANALYSIS

Highest satisfaction score is Jurong East Street, Lowest is Tampines - Target the lowest 5 branches for retraining

44.5 % of all interactions are simple balance enquiries - Immediate candidate for self-service migration

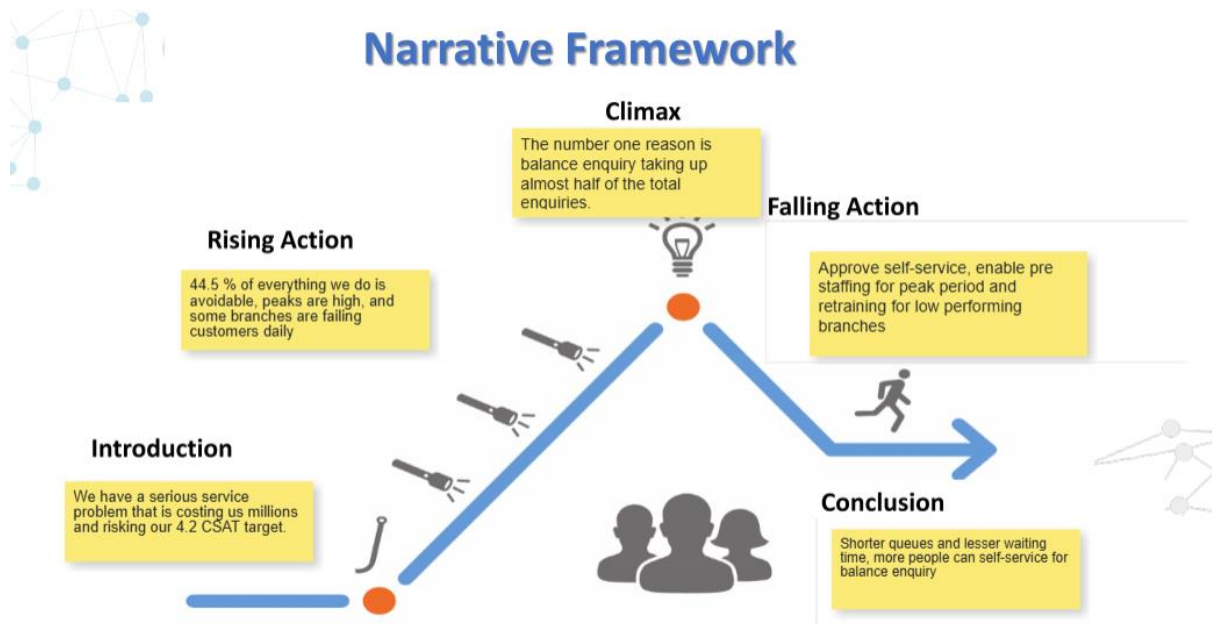
The peak period for branch is in August 2023 while for Phone it is October 2024 - Pre-schedule 20-25 % extra staff during peak months

OUTCOME

Approve : Digital balance migration , peak staffing , branch retraining

Shorter queues and lesser waiting time, more people can self service for balance enquiry

Narrative framework using Freytag's Pyramid



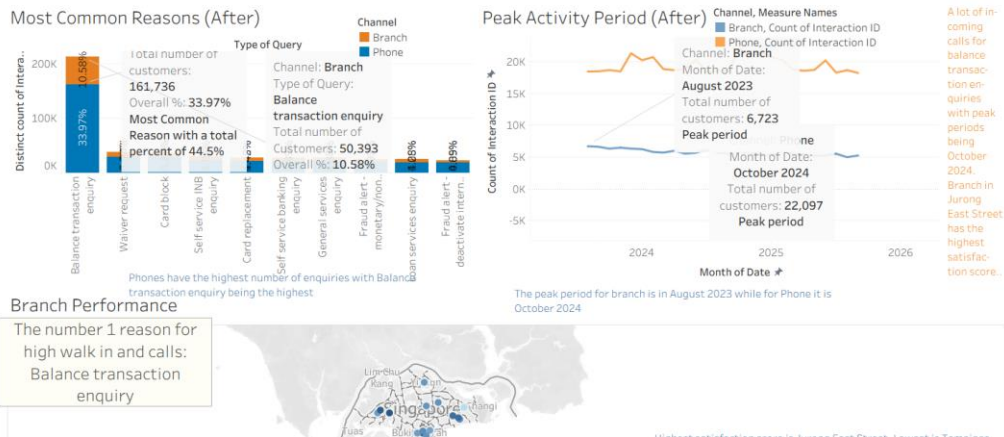
Task 5 : Tableau Story

I have used the narrative framework using Freytag's Pyramid for my Tableau Story. My first point shows the full dashboard to provide the readers with an overall introduction. Rising action was my points 2-5 to show tension building with multiple evidence such as phones overload, spikes, branch CSAT (Customer Satisfaction Score) and long waiting times. For the climax, I talked about the 44.5% of all enquiries being balance transaction and highlighted the bar. For falling action, I moved more towards the solution to approve the 3 main points. Finally, my conclusion highlighted the problems and recommendation.

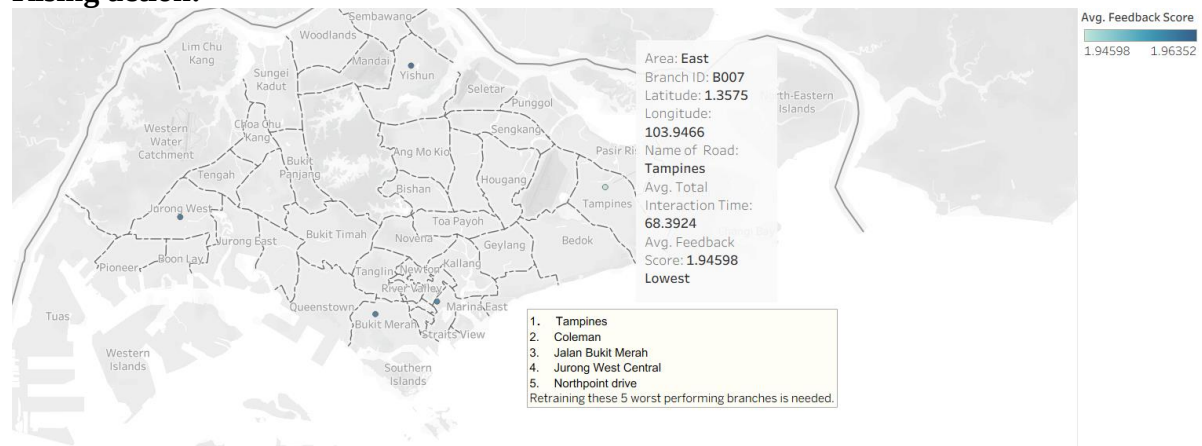
My various visuals fit well with my story points. The stacked bar instantly highlights the query type and the one that stands out. Dual axis line will show the seasonal spikes which can be used for the upcoming years. Map tells us the location and the feedback score, so it guides us on which branches are performing poorly, especially targeting the lowest 5 for retraining. My scatter plot for feedback and waiting time tells us about when customers wait too long their experience becomes worse leading to a lower feedback score. Also, my simple bar chart of channel vs feedback supports the story point by showing us how poorly branches are performing as compared to phones which shows the importance of needing to retrain branches.

My 3 recommendations are to improvise digital self-service migration, add seasonal staffing and a targeted retraining programme. All 3 recommendations fit the problem statement to manage the volume of inquiries. These will overall reduce a customer's waiting time; ensure they have enough staff for the customers to attend and ensure branches especially give a better-quality service to better satisfy the customers and achieve a higher CSAT score.

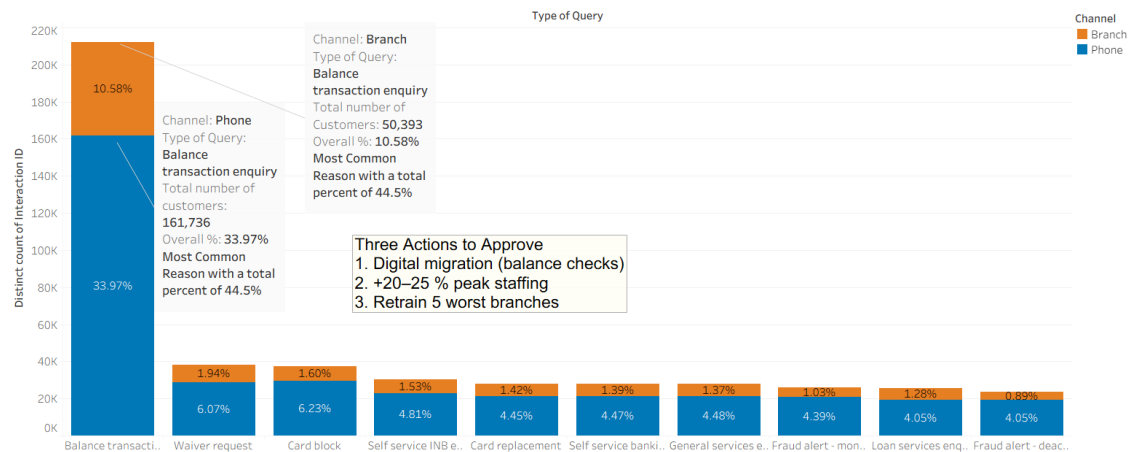
Hook:



Rising action:



Actions to take:



Recommendation:

Conclusion:

High Balance Transaction enquiries
Branch has a very low average feedback score
Peak periods are August 2023, October 2024

Recommendation:

Approve the plan for self-service,
peak-staffing
Achieve the 4.2 CSAT target

Task 6 : Gen AI Reflection Journal

Gen AI overall produced a relevant and good actionable recommendation. For visualisation techniques, it suggested charts such as bar chart for channel and map for latitude longitude, which I implemented to highlight the 44.5% balance enquiries and Tampines branch's lowest feedback score. It had even gone the extra mile to suggest a few multi variable insights such as using a box plot or violin plot for channel vs waiting time. It did not only suggest one chart for each variable but suggested 2-3. This gave me a shorter range of charts to choose from saving time. I implemented the stacked bar chart for query types which helped me reveal the phone channel's dominance in most of the queries, especially the balance transaction. However, I feel that it could have done better by suggesting one appropriate chart which is the best first, then proceed to tell me alternatives. I was able to use the most relevant charts according to the problem statement and produce good visualisation with the appropriate type of chart.

As for the dashboard design, one important thing it helped me with is the layout and the use of white space. A 50-50 split at the top half and a middle row + bottom row layout allowed me to show my visualisations clearly without any overlapping. There was also enough white space available for me to put in my annotations. The option to allow floating filters was also useful as I was able to make use of the white space once again. I was able to implement a channel floating filter and was also able to use my graphs as a filter. For example, when I click on the phone channel, it will only show the plots with phone, allowing it to filter for easier specific visualisations.

Suggestions for storyboarding were in depth and useful, it gave a 6-step visual narrative with introduction based on the context and even provided 7 recommendations such as virtual queues to reduce walk in traffic. With the information that Gen AI gave me, I was able to expand my story into a 10-point narrative in Tableau with the Freytag's Pyramid that I had created earlier. One thing that I incorporated was the digital migration for 44.5% balance enquiries like what the AI suggested. Another one was to improve branch staffing during peak hours. This fit my story point well as I had a graph showing peak activity period which

highlighted the months where there were a lot of customers walking in and calling for enquiries. Increasing the staffing can allow more customers to be attend to.

Overall, Gen AI was able to respond well and give me good suggestions which helped me save some time on deciding what to do. With proper prompts, future use can be improved if we give it more information about the dataset. Gen AI helped me improve my charts, dashboard and storyboarding.